Monitoring and Operations in AWS Lesson

Skills you will learn

Cloud Computing: Monitoring & Scaling

Lesson Overview

AWS provides a suite of monitoring and operational tools that help manage system performance, security, and event-driven automation. This lesson will introduce key AWS monitoring services, including Amazon CloudWatch, AWS CloudTrail, and Amazon EventBridge.

Learning Objectives

By the end of this lesson, you will be able to:

1. Understand Amazon CloudWatch and how it collects and visualizes real-time metrics.
2. Configure and manage CloudWatch Alarms to automate responses based on monitored thresholds.
3. Explore AWS CloudTrail and its role in logging and monitoring AWS API calls.
4. Utilize logs for security and compliance to analyze user activities and system interactions.
5. Understand Amazon EventBridge for building event-driven architectures across AWS and third-party applications.

Monitoring and Operations in AWS

1. Amazon CloudWatch

Amazon CloudWatch is a real-time monitoring service that collects and analyzes performance data from AWS services, applications, and system resources. It provides visibility into operational health and enables proactive adjustments to enhance system efficiency.

Key Features of CloudWatch:

- **Metrics:** CloudWatch tracks system metrics like CPU utilization, memory usage, and disk I/O.
- **Dashboards:** Users can visualize data trends and gain insights into system perform
- **Logs:** Stores system logs for debugging and analysis.

Ask AI Tutor

- **Alarms:** Triggers notifications or automated responses based on monitored thresholds.

📌 **Example:** Monitoring an EC2 instance's CPU utilization and automatically scaling up resources if CPU usage exceeds 80% for more than 5 minutes.

2. Amazon CloudWatch Alarms

CloudWatch Alarms monitor system metrics and take predefined actions when thresholds are breached. They help automate responses such as sending alerts, stopping instances, or triggering AWS Lambda functions.

CloudWatch Alarm States:

1. **OK** – The metric is within the defined threshold.
2. **ALARM** – The metric exceeds the set threshold.
3. **INSUFFICIENT DATA** – Not enough data is available to determine the metric state.

📌 **Example:** Setting up a CloudWatch Alarm that sends an SNS notification when available EC2 instance memory drops below 500MB.

3. AWS CloudTrail

AWS CloudTrail is a logging service that records API activity across AWS services. It helps in governance, security analysis, and compliance tracking.

Key Features of CloudTrail:

- **Records AWS API calls** made via the console, CLI, or SDK.
- **Stores logs in Amazon S3** for future analysis.
- **Integrates with AWS Athena** to query and analyze logs.

CloudTrail logs include: ✅ **Who made the request?** ✅ **What actions were performed?** ✅ **When did the request occur?** ✅ **What was the response?**

📌 **Example:** A company using CloudTrail logs to investigate unauthorized access attempts in their AWS account.

4. Amazon EventBridge

Amazon EventBridge enables event-driven automation by connecting AWS services and third-party applications. It allows you to define rules that respond to specific events in real time.

Key Features of EventBridge:

- **Event Routing:** Automatically triggers actions based on events (e.g., Lambda functions, SNS notifications).
- **AWS Service Integration:** Works seamlessly with CloudWatch, S3, EC2, and more.
- **Third-party Integration:** Connects with external applications like Zendesk, Datadog, or PagerDuty.

📌 **Example:** Using EventBridge to trigger an AWS Lambda function whenever an S3 bucket receives a new file upload.

Summary

- **CloudWatch** provides real-time monitoring and alarms for AWS resources.
- **CloudTrail** logs API activities for auditing and security analysis.
- **EventBridge** enables event-driven workflows by routing system events to AWS services.

These tools together help in optimizing AWS environments, improving system performance, and ensuring compliance.

AI TUTOR ACTIONS

Generate a practice quiz Summarize this unit Ask about this unit

Complete unit

Previous

Week 2: Day 4
Elastic Load Balancer & Auto Scaling in AWS

Next